Glaze Search · Quick Start

For new users: install once, then follow one example click by click, from a photo of a pot to the glaze recipes that could make it. Every screenshot below is the real app doing exactly these steps (6 November 2026); if you do the same, you see the same.

1 · Install once (or use it online)

No install: open `https://huggingface.co/spaces/FBIWarninggg/glaze_search` and log in to Hugging Face (more free GPU time). **On your computer:** clone the branch, then run five launchers in order (Mac: the `.command` files, Apple Silicon M1 or newer).

```
git clone -b nghia_mike_ceramic_search --single-branch https://github.com/haohaozai/GC2-Ensemble-Model-Pottery.git
```

#	WINDOWS	MACOS	WHAT IT DOES	WHEN
1	`SETUP.bat`	`SETUP.command`	Python environment and the pretrained models (about 3.7 GB); needs `uv` once	once
2	`DOWNLOAD_DATA_ORIGINAL_AND_SEGMENTATION` .bat / .command		photos, cut-outs and search index (8 archives, about 6.15 GB); run again if it stops	once
3	`CHECK_DATA.bat`	`CHECK_DATA.command`	every line should say OK	once
4	`INSTALL_GLAZE_MODEL.bat`	`INSTALL_GLAZE_MODEL.command`	the recipe models from the team's `test_similarity.zip` (drag it onto the .bat; Mac: put it in Downloads)	once
5	`RUN_GLAZE_SEARCH.bat`	`RUN_GLAZE_SEARCH.command`	starts the app at `http://127.0.0.1:8770` ; keep the window open	every time

2 · The example, click by click

The task: you have a photo of a glossy blue bowl (held in your hand) and fire to cone 6 in an electric kiln (oxidation). Which Glazy recipe makes this glaze?

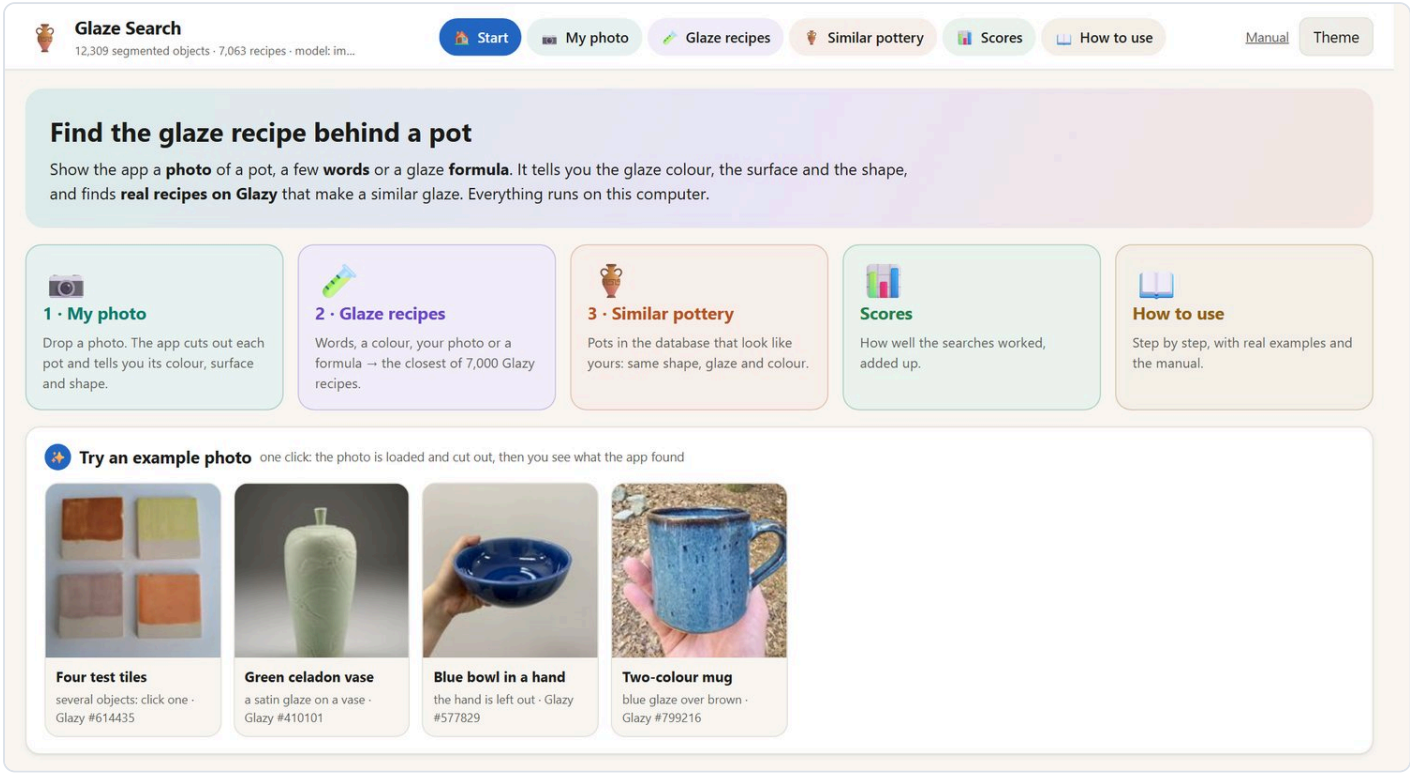
1 Start page: load the photo

DO

Under **Try an example photo** click **Blue bowl in a hand**. (With your own pot: click 📷 **My photo** and drop, paste or browse the photo.)

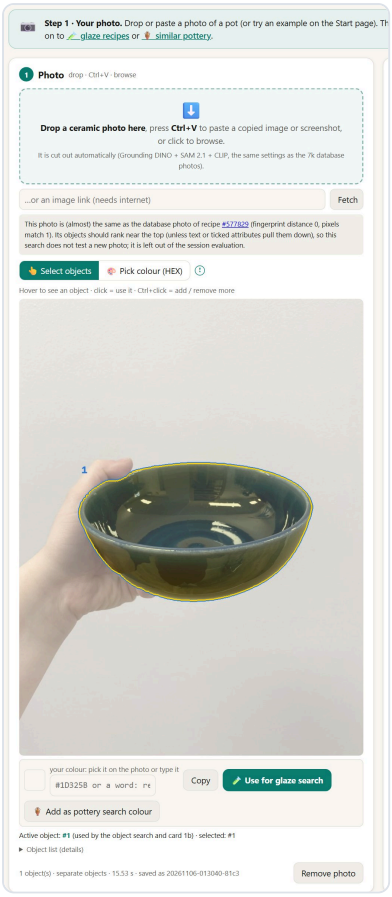
THEN

The app opens 📷 **My photo** and cuts the pot out: a few seconds with an NVIDIA GPU, about a minute on a CPU or a Mac.



The 🏠 Start page. The third example photo is the one used here.

2 My photo: the pot is found and cut out



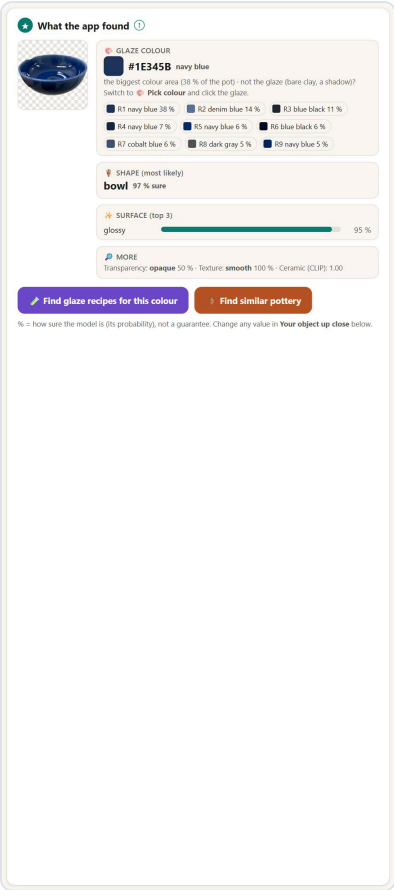
📷 My photo after the example loaded (the photo half of the page).

You see: the bowl outlined and shaded (the shading marks the object in use); your hand is **not** part of the object.

The grey note says this photo is already in the Glazy database (recipe #577829). That is expected for an example photo: it is left out of the session scores.

Diagram: A1 Grounding DINO finds the bowl · A2 SAM 2.1 traces its outline · A3 CLIP checks it is ceramic

3 Read “What the app found”



What the app found, on the right of the photo page.

PANEL	SHOWS
🎨 Glaze colour	**#1E345B navy blue**: the biggest colour area (38 % of the pot), then the other colour regions
🏺 Shape	**bowl**, 97 % sure
✨ Surface	**glossy**, 95 %
🔎 More	opaque 50 % · smooth 100 % · ceramic (CLIP) 1.00

Check the colour is the **glaze**, not bare clay or a shadow. If not: 🎨 **Pick colour** and click on the glaze.

Diagram: A4 colours (ColorThief + CIELAB) · A5 six attributes

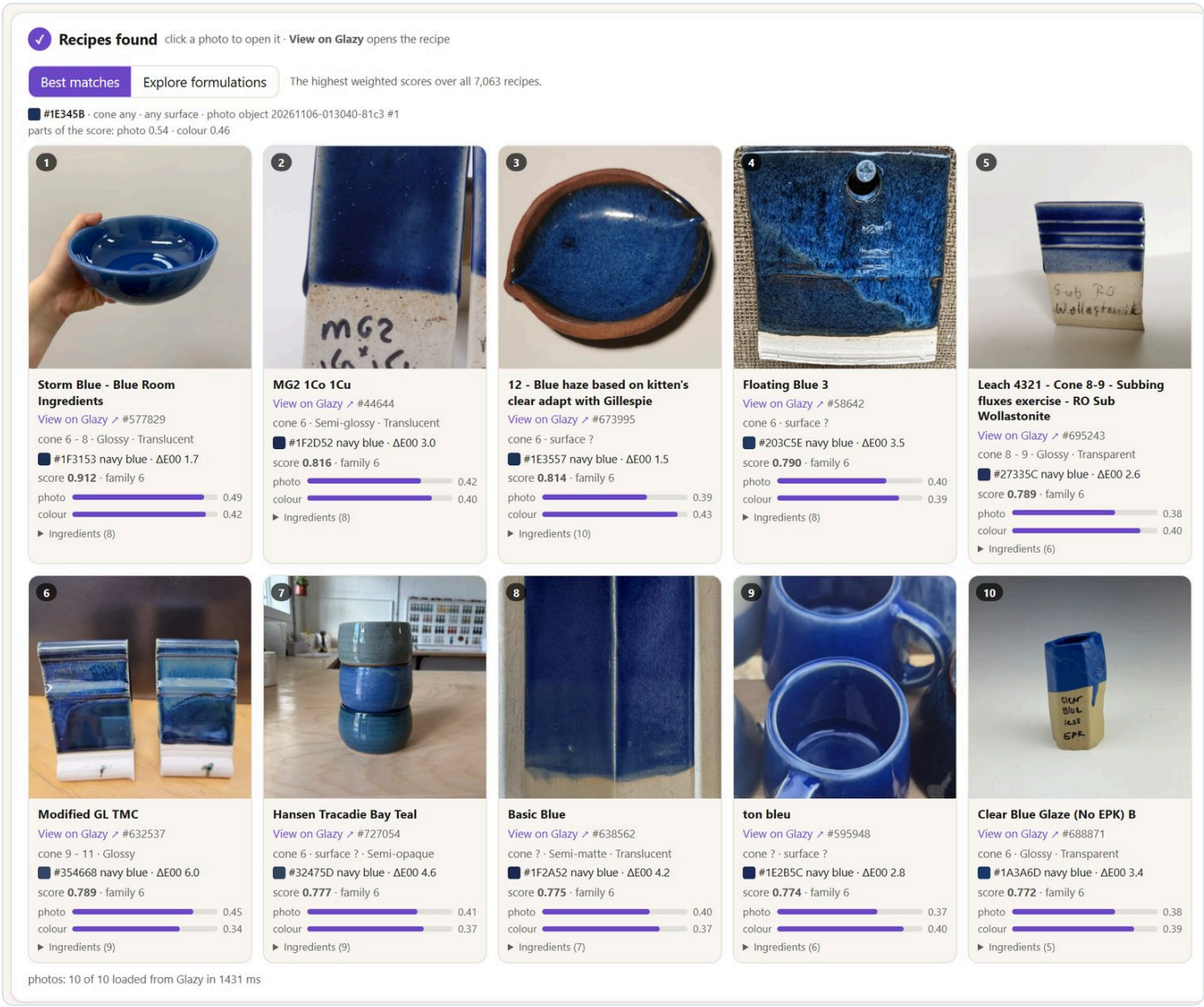
4 Press 🧪 Find glaze recipes for this colour

DO

Press the purple button at the bottom of **What the app found**.

YOU SEE

The 🧪 **Glaze recipes** page, already searched with your photo and its colour: *photo 0.54 · colour 0.46*.



The first result: #1 Storm Blue - Blue Room Ingredients (score 0.912), #2 MG2 1Co 1Cu (0.816), #3 12 - Blue haze based on kitten's clear adapt with Gillespie (0.814).

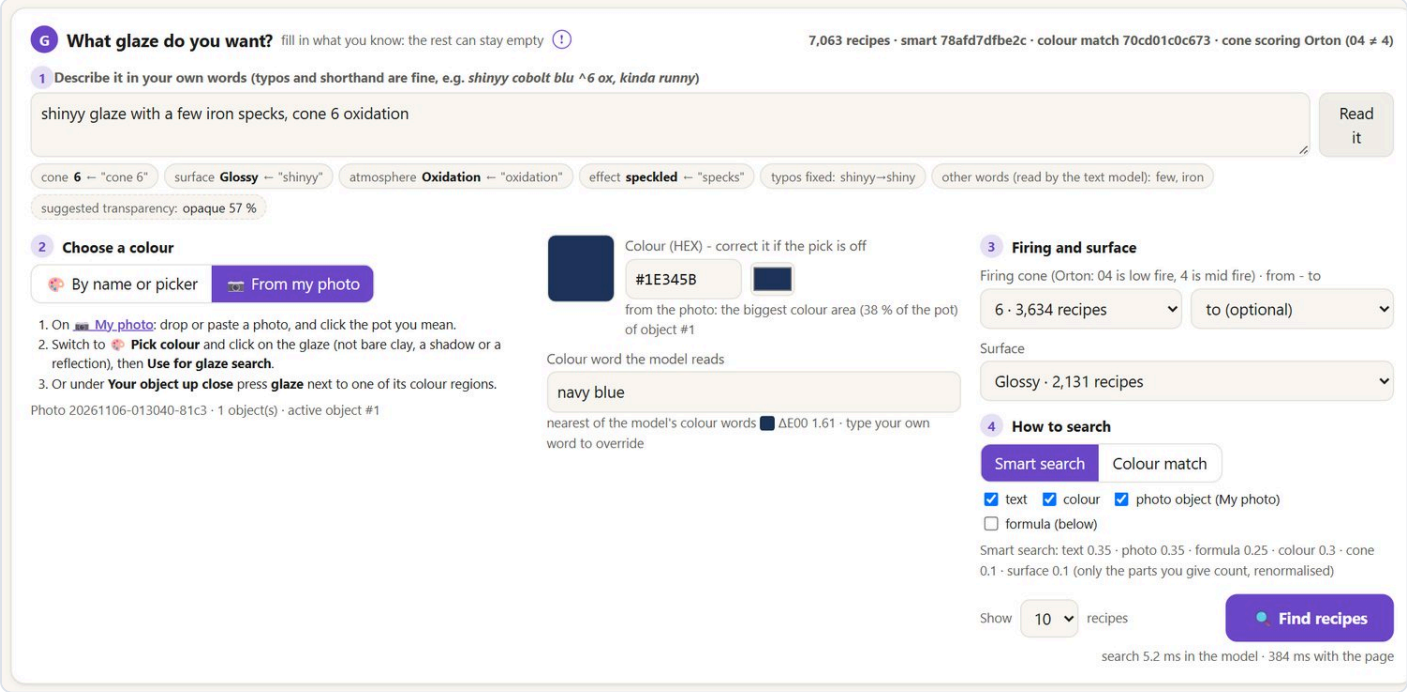
5 Describe what you want, in your own words

DO

Scroll up to **1 · Describe it**, type `shinyy glaze with a few iron specks, cone 6 oxidation` and press **Read it**. Typos are fine.

YOU SEE

Chips with what it understood: **cone 6** ← "cone 6" · **Glossy** ← "shinyy" · **Oxidation** · effect **speckled** ← "specks" · typos fixed: shinyy → shiny. Cone and surface are filled in under 3.

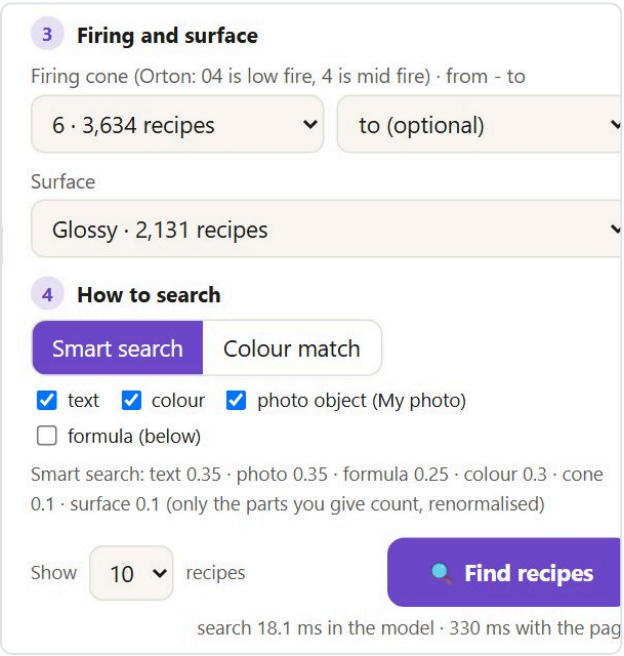


Card G after Read it: the chips under the sentence; the colour is still the photo's #1E345B ("from the photo: the biggest colour area").

> **Do not name the colour in the words when you use a photo.** If the sentence says a colour (for example "dark blue"), Read it puts *that* colour into Colour (HEX) and the photo's colour is no longer used. If it happens, type the photo's HEX (here `#1E345B`) back into Colour (HEX).

Diagram: B3 read the words (rules + fuzzy matching) · B2 MiniLM reads the whole sentence

6 Check firing and engine, then search



3 · Firing and surface and 4 · How to search.

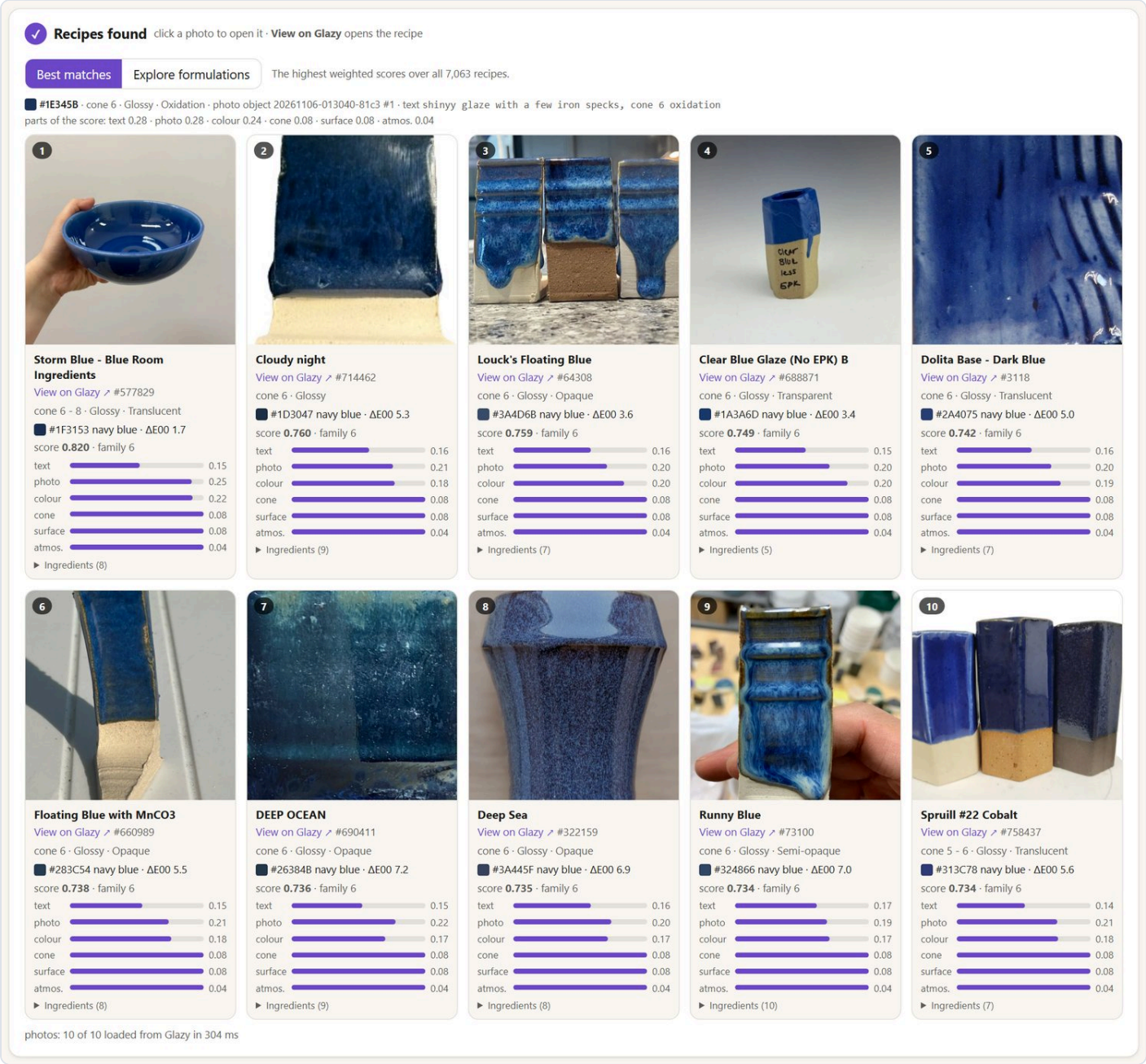
3 · Firing and surface: cone 6 and Glossy, from your words (cone **04** is low fire, not cone 4).

4 · How to search: Smart search, with text, colour and photo object ticked. **Show 10**.

Do: press 🔍 **Find recipes**. It takes under a second.

Diagram: join → C2 smart search (3 trained heads on frozen MiniLM and DINOv2) → rank every recipe

7 The result: Best matches

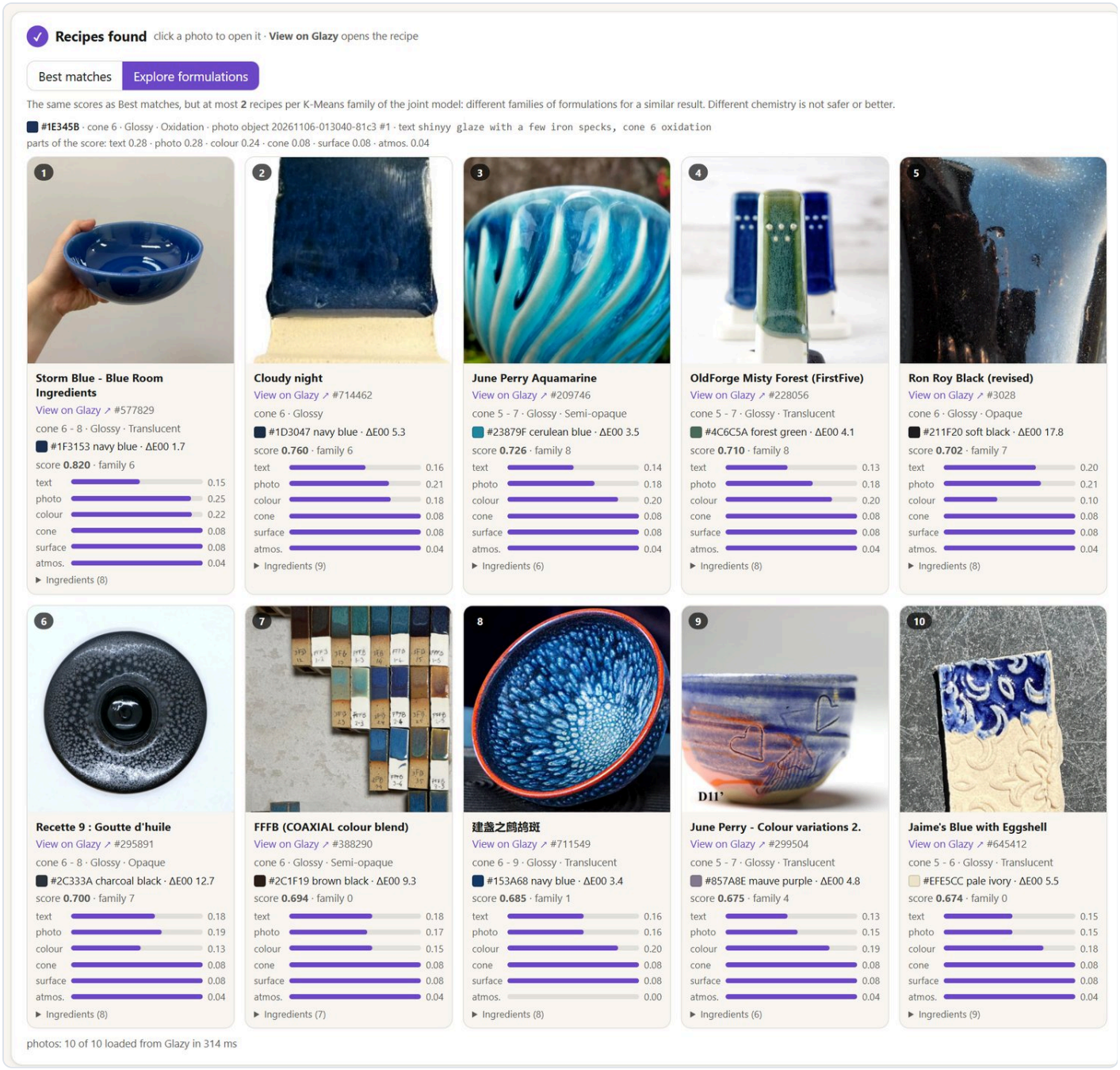


Best matches, scored over all 7,063 recipes with *text 0.28 · photo 0.28 · colour 0.24 · cone 0.08 · surface 0.08 · atmos. 0.04*. Every photo comes from Glazy.

> **How to read a card:** the Glazy photo and name (**View on Glazy ↗** opens the recipe); cone, surface and transparency; the colour with its **ΔE00** (below about 5 the colours look alike, above 10 they are different colours); the score, with bars for what each part added.

8 Explore formulations

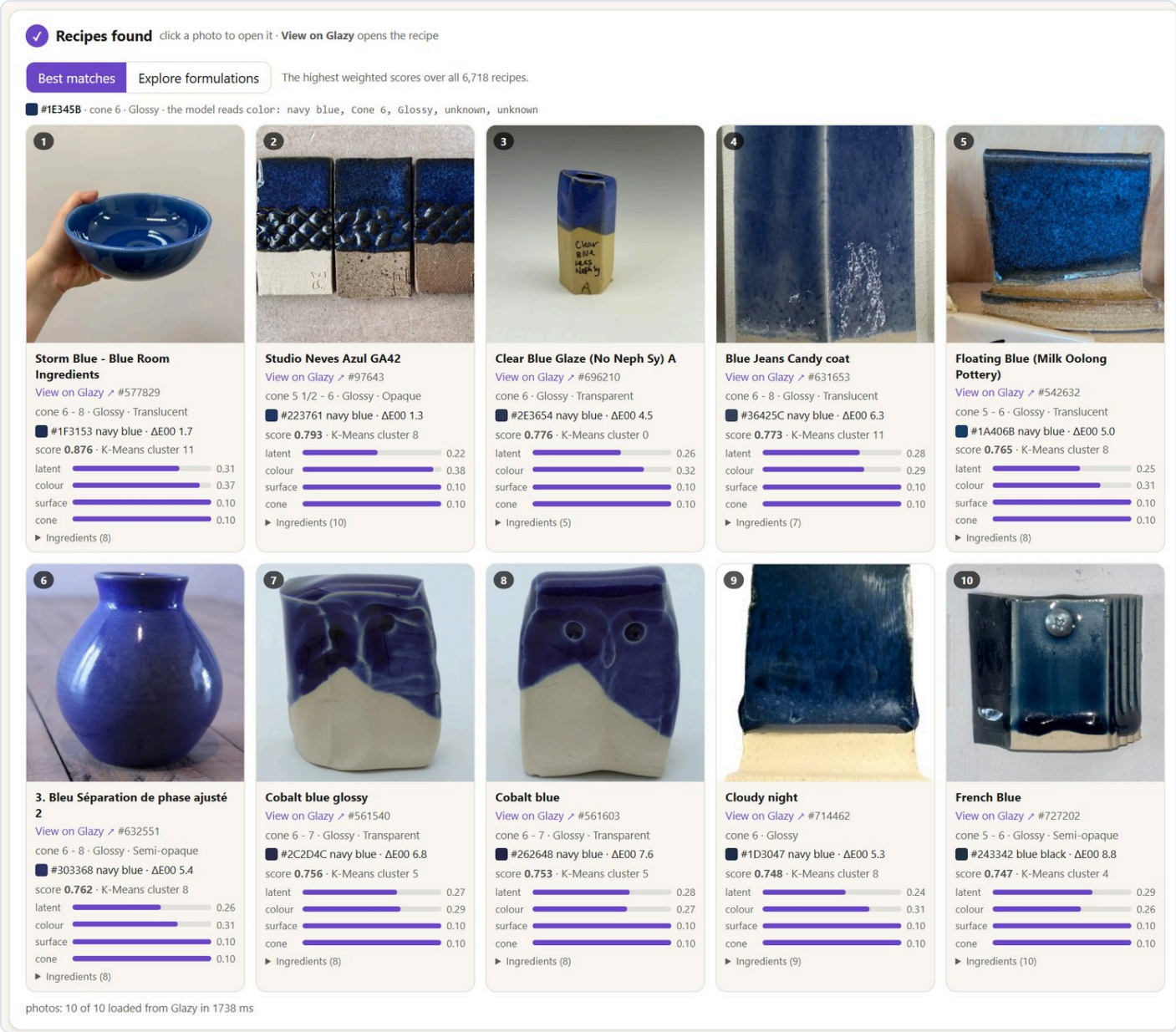
Click the **Explore formulations** tab: the same scores, but at most 2 recipes per chemistry family, so you see different bases for a similar look.



Explore formulations: Storm Blue - Blue Room Ingredients · Cloudy night · June Perry Aquamarine · OldForge Misty Forest (FirstFive) · Ron Roy Black (revised) …

9 (Optional) Ask the second engine: Colour match

Click **Best matches**, switch **4 · How to search** to **Colour match** and press 🔍 **Find recipes**. It reads only colour + cone + surface (`color: navy blue, Cone 6, Glossy, unknown, unknown`) and compares them with each recipe's chemistry: 0.40 latent (learned) match + 0.40 colour + 0.10 surface + 0.10 cone.



Colour match: #1 Storm Blue - Blue Room Ingredients (0.876), #2 Studio Neves Azul GA42 (0.793), #3 Clear Blue Glaze (No Neph Sy) A (0.776).

10 Decide

Open the top cards with **View on Glazy ↗** for the ingredients. Prefer a small ΔE00 and the right cone for your kiln, then **test-fire a small tile** before you mix a big batch.

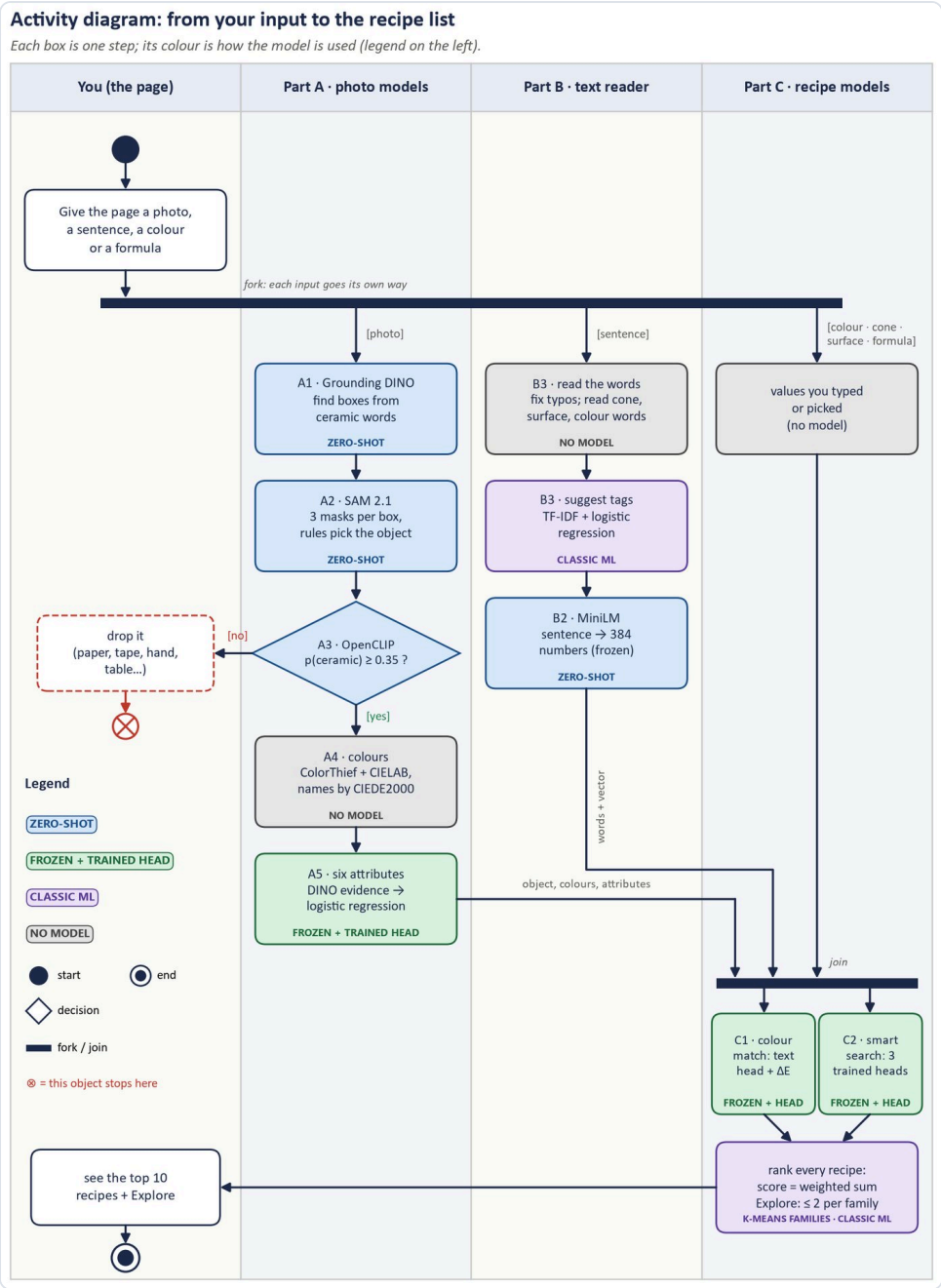
The answer: Storm Blue - Blue Room Ingredients (Glazy #577829, #1F3153 navy blue · ΔE00 1.7), first in Smart search (score 0.820) and in Colour match (0.876).
It is this bowl's own recipe: the example photo comes from the Glazy database, so the app found the exact recipe from the photo and a sentence with a typo. With your own pot there is no “exact” recipe: the list gives the closest glazes (here Cloudy night, Louck's Floating Blue …), and you choose with ΔE00 and the cone.

3 · Your results at a glance

STEP	SEARCH	TOP 3 (SCORE)
4	photo + colour	Storm Blue - Blue Room Ingredients (0.912) · MG2 1Co 1Cu (0.816) · 12 - Blue haze based on kitten's clear adapt with Gillespie (0.814)
7	Smart search: words + photo + colour + cone + surface + atmosphere	Storm Blue - Blue Room Ingredients (0.820) · Cloudy night (0.760) · Louck's Floating Blue (0.759)
9	Colour match: colour + cone + surface	Storm Blue - Blue Room Ingredients (0.876) · Studio Neves Azul GA42 (0.793) · Clear Blue Glaze (No Neph Sy) A (0.776)

4 · What happened behind each step

STEP	BOX IN THE ACTIVITY DIAGRAM	HOW THE MODEL IS USED
2	A1 Grounding DINO · A2 SAM 2.1 · A3 CLIP	pretrained, zero-shot: a text prompt, a box prompt, 30 sentences
3	A4 colours · A5 six attributes	ColorThief + CIELAB (no model) · DINO evidence → small trained layer
5	B3 read the words · B2 MiniLM	rules + fuzzy matching · frozen text model
6–7	join → C2 smart search → rank	3 small trained heads on frozen MiniLM and DINOv2
8	Explore: ≤ 2 per K-Means family	K-Means families of the joint model
9	C1 colour match	frozen MiniLM + 2 trained layers (text ↔ chemistry)



The activity diagram of one search.

5 · Problems

IF…	THEN…
the colour is the bare clay or a shadow	📷 My photo → 🎨 Pick colour and click on the glaze
the colour changed after Read it	your words named a colour: type the photo's HEX back into Colour (HEX), or leave the colour out of the words
the wrong pot is used	click the pot you mean on 📷 My photo
online: “Hugging Face's free GPU: RuntimeError”	try again in a minute; the Space owner can restart the Space
online: “free GPU limit”	log in to Hugging Face, or wait: the allowance renews

More: the Manual (all six pages, six worked examples), Process and models, Pretrained models, Evaluation, in `docs/manuals`.

Glaze Search Quick Start · screenshots and numbers from the app on 6 November 2026 (Glaze Search server with the installed models: colour match 70cd01c0c673, smart search 78afd7dfbe2c). Recipe photos and names belong to their authors on glazy.org.